



Mitoprep Test 2

Date – 10 December 2025

Duration – 3 Hours

Marks – 720

Physics :- Laws of Motion, Work, Energy & Power

Chemistry :- Equilibrium

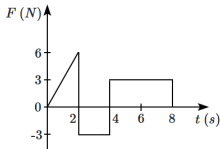
Biology :- Biological Classification, Plant Kingdom, Morphology of Flowering Plants,
Body Fluids & Circulation, Excretory Products & their Elimination

Instruction

1. The test is of 3 hours duration.
2. The test booklet consists of 200 questions. The maximum mark is 720.
3. There are four Sections in the Question Paper, Sections I, II, III, and IV consisting of Section I (Physics), Section II (Chemistry), Section III (Botany) and Section IV (Zoology) have 50 Questions in each Subject and each subject is divided into two Sections,
4. Section A consists of 35 questions (all questions compulsory) and Section B consists of 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed, or written, bits of paper, pager, mobile phone, any electronic device, etc. Inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the Candidates are allowed to take away this Test Booklet with them

Physics

1. The force F acting on a particle of mass m is indicated by the force-time graph shown below. The change in momentum of the particle over the time interval from 0 to 8 s is:



1. 24 N-s
2. 20 N-s
3. 12 N-s
4. 6 N-s

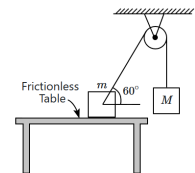
2. The energy that will be ideally radiated by a 100 kW transmitter in 1 hr is:

1. 1×10^5 J	2. 36×10^7 J
3. 36×10^4 J	4. 36×10^5 J

3. A 0.5 kg ball traveling at 10 m/s collides with a stationary 2.0 kg ball and rebounds in the opposite direction at 6 m/s. What is true, from the options given below, for the speed of the 2.0 kg ball after collision?

1. The 2.0 kg ball moves at 10 m/s.
2. The 2.0 kg ball moves at 6 m/s.
3. The 2.0 kg ball moves at 4 m/s.
4. The 2.0 kg ball moves at 2.5 m/s.

4. What is the minimum value of the mass M so that the block is lifted off the table at the instant shown in the diagram? Assume that the blocks are initially at rest.



1. $\frac{m}{\sin 60^\circ}$	2. $\frac{m}{\tan 60^\circ}$
3. $m \sin 60^\circ$	4. $\frac{m}{\tan 30^\circ}$

5. A 5 kg stationary bomb explodes in three parts with masses in the ratio 1 : 1 : 3 respectively. If parts having the same mass move in perpendicular directions with velocity 30 m/s, then the speed of the bigger part will be:

1. $10\sqrt{2}$ m/s	2. $\frac{10}{\sqrt{2}}$ m/s
3. $13\sqrt{2}$ m/s	4. $\frac{15}{\sqrt{2}}$ m/s

6. A car with a mass of 200 kg is moving along a circular track with a radius of 70 m at an angular velocity of 0.2 rad/s. What is the magnitude of the centripetal force acting on the car?

1. 560 N
2. 400 N
3. 360 N
4. 200 N

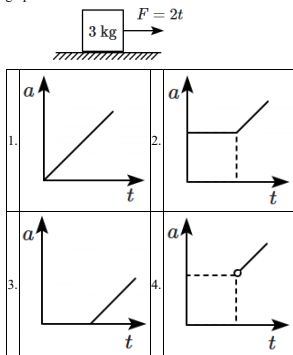
7. If the normal force is doubled, the coefficient of friction is:

1. not changed	2. halved
3. doubled	4. tripled

8. A particle moves with a velocity $(5\hat{i} - 3\hat{j} + 6\hat{k}) \text{ ms}^{-1}$ horizontally under the action of a constant force $(10\hat{i} + 10\hat{j} + 20\hat{k}) \text{ N}$. The instantaneous power supplied to the particle is:

1. 200 W	2. zero
3. 100 W	4. 140 W

9. A block of mass 3 kg is placed on a rough surface ($\mu = 0.2$) and a variable force acts on it. Variation of acceleration of block with time is correctly shown by the graph:



10. Two spheres A and B of masses m_1 and m_2 , respectively, collide. A is at rest initially and B is moving with velocity v along the x -axis. After the collision, B has a velocity $\frac{v}{2}$ in a direction perpendicular to the original direction. The mass A moves after collision in the direction:

1. same as that of B .
2. opposite to that of B .
3. $\theta = \tan^{-1} \left(\frac{1}{2} \right)$ to the positive x -axis.
4. $\theta = \tan^{-1} \left(\frac{-1}{2} \right)$ to the positive x -axis.

11. A bird is sitting in a wire cage hanging from the spring balance. Let the reading of the spring balance be W_1 . If the bird flies about inside the cage, the reading of the spring balance is W_2 . Which of the following is true?

- $W_1 = W_2$
- $W_1 > W_2$
- $W_1 < W_2$
- Nothing definite can be predicted

12. What tension must a rope exert to support a precious painting with a mass, m on a smooth, frictionless surface inclined at an angle, θ with respect to the horizontal ground?

- mg
- $mg \sin \theta$
- $mg \cos \theta$
- $mg \tan \theta$

13. If a motor, that generates 25 kJ in 4 min and 10 s, increases its power output by 50%, then the new power output of the motor is:

1. 25 W	2. 50 W
3. 100 W	4. 150 W

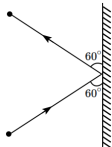
14. In a certain physics experiment, two carts run along a frictionless, level, one-dimensional track. The mass of cart A is 1 kg and the mass of the cart B is 2 kg. Initially cart A is moving to the right at 0.2 m/s, and cart B is moving to the left at 0.7 m/s (see figure). After they collide, they stick together.



What is the magnitude of the total momentum of the system before the collision?

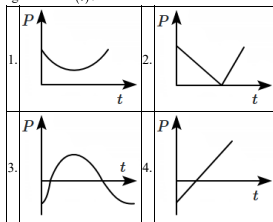
- 1.2 kg-m/s
- 1.5 kg-m/s
- 1.6 kg-m/s
- 2.7 kg-m/s

15. A 3 kg ball strikes a heavy rigid wall with a speed of 10 ms^{-1} at an angle of 60° . It gets reflected with the same speed and angle as shown in the figure. If the ball is in contact with the wall for 0.20 s, what is the average force exerted on the ball by the wall?



1. 300 N
2. 150 N
3. zero
4. $150\sqrt{3} \text{ N}$

16. A particle is projected at a time $t = 0$ with a speed v_0 and at an angle with the horizontal in a uniform gravitational field. Then which of the following graphs represents power delivered by the gravitational force against time (t)?



17. Given below are two statements:

Assertion (A):	When a firecracker (rocket) explodes in mid-air, its fragments fly in such a way that they continue moving in the same path, which the firecracker would have followed, had it not exploded.
Reason (R):	The explosion of cracker (rocket) occurs due to internal forces only and no external force acts for this explosion.

1.	Both (A) and (R) are True and (R) is the correct explanation of (A).
2.	Both (A) and (R) are True but (R) is not the correct explanation of (A).
3.	(A) is True but (R) is False.
4.	(A) is False but (R) is True.

18. How many 2.5 kg bricks can a man carry up a staircase 3.6 m high in one hour if he works at the average rate of 9.8 watts?

1.	800	2.	200
3.	600	4.	400

19. Given below are two statements:

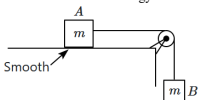
Statement I:	Work done by all forces on a body equals the change in its kinetic energy.
Statement II:	Work done by non-conservative forces does not change the kinetic energy of a system.

1.	Statement I is incorrect and Statement II is correct.
2.	Both Statement I and Statement II are correct.
3.	Both Statement I and Statement II are incorrect.
4.	Statement I is correct and Statement II is incorrect.

20. A block of mass 2 kg is sliding on a rough horizontal surface with an initial velocity 20 m/s. The block comes to rest after traveling a distance of 40 m. The value of the coefficient of kinetic friction acting between the block and surface is: (take $g = 10 \text{ m/s}^2$)

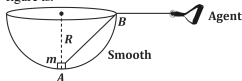
1. 0.2
2. 0.25
3. 0.4
4. 0.5

21. The system, shown in adjacent diagram, is released from rest. The block B falls through a height h . There is no friction anywhere, and the pulley and string are ideal. The increase in the kinetic energy of A is:



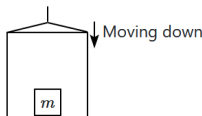
1. zero	2. mgh
3. $\frac{mgh}{2}$	4. $2mgh$

22. The minimum work done by the agent in pulling a small particle of mass m from A to B as shown in the figure is:



- $4mgR$
- mgR
- $3mgR$
- $2mgR$

23. A box of mass m is placed in an elevator (lift), which is moving downward at constant velocity. Let the work done by the normal reaction (N) of the "ground" on the box be W_N , and the work done by gravity on the box be W_g in the same time. Then, which statement is true in the ground frame as well as the elevator's frame?



- $W_N = 0, W_g > 0$
- $W_N > 0, W_g = 0$
- $W_N = W_g = 0$
- $W_N + W_g = 0$

24. Identify the correct statements:

(A)	Every force encountered in mechanics has an associated potential energy.
(B)	The work done by all the forces is equal to the change in kinetic energy.
(C)	The work done by all the forces is equal to the change in potential energy.
(D)	The total mechanical energy of a system is conserved if the forces, doing work on it, are conservative.

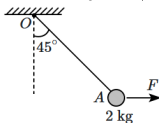
Choose the correct option from the options given below:

1. (A) and (B) only
2. (C) and (B) only
3. (C) only
4. (B) and (D) only

25. A box of mass M lies on the floor of a descending lift. The lift accelerates downward with an acceleration a . What should be the value of a such that the box exerts a force of $\frac{Mg}{4}$ on the floor?

1. $\frac{g}{4}$	2. $\frac{g}{2}$
3. $\frac{3g}{4}$	4. $3g$

26. A simple pendulum consists of a bob (A) of mass 2 kg connected by means of a light, inextensible string of length 1 m to a fixed point O on the ceiling. A horizontal force is applied to the bob (A); the string makes an angle of 45° with the vertical & the system is stationary. Take $g = 10\text{ m/s}^2$, if required.



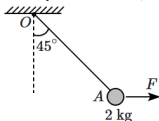
The magnitude of the tension in the string is:

- 10 N
- 20 N
- $10\sqrt{2}\text{ N}$
- $20\sqrt{2}\text{ N}$

27. A cyclist speeding at 5 ms^{-1} on a level road takes a sharp circular turn of radius 5 m without tilting and reducing the speed. What should be the minimum value of the coefficient of static friction between the tyres and the road so that the cyclist will not slip while taking the turn? (take $g = 10 \text{ ms}^{-2}$)

1. 0.50
2. 0.25
3. 0.20
4. 0.15

28. A simple pendulum consists of a bob (A) of mass 2 kg connected by means of a light, inextensible string of length 1 m to a fixed point O on the ceiling. A horizontal force is applied to the bob (A); the string makes an angle of 45° with the vertical & the system is stationary. Take $g = 10 \text{ m/s}^2$, if required.



The magnitude of F is:

1. 10 N
2. 20 N
3. $10\sqrt{2}$ N
4. $20\sqrt{2}$ N

29. A block of mass 10 kg is in contact with the inner wall of a hollow cylindrical drum of radius 1 m. The coefficient of friction between the block and the inner wall of the cylinder is 0.1. The minimum angular velocity needed for the cylinder, which is vertical and rotating about its axis, will be:

- ($g = 10 \text{ m/s}^2$)
1. $10 \pi \text{ rad/s}$
 2. $\sqrt{10} \pi \text{ rad/s}$
 3. $\frac{10}{\pi} \text{ rad/s}$
 4. 10 rad/s

30. An electric train is traveling along a straight, horizontal track. A constant resultant force (greater than zero) acts on the train in the direction of its motion. What happens to the magnitude of the train's acceleration and velocity while this force is acting?

1.	The acceleration increases, and the velocity remains constant.
2.	The acceleration remains constant, and the velocity increases.
3.	The acceleration decreases, and the velocity increases.
4.	The acceleration remains constant, and the velocity remains constant.

31. A man (mass : m) climbs a staircase of height h , slowly. The work done on the man by the reaction force exerted by the stairs is:

1. mgh
2. $\frac{mgh}{2}$
3. $-mgh$
4. zero

32. A cyclist comes to a skidding stop over a distance of 10 m. During the process, the force on the cycle due to the road is 200 N and directly opposed to the motion. How much work does the road do on the cycle?

1. zero
2. 2000 J
3. -2000 J
4. -200 J

33. An astronaut, in a space shuttle, orbiting close to the earth's surface (take $g = 10 \text{ m/s}^2$), suddenly fires his engines so as to give him a forward acceleration of $\frac{3g}{4}$ along the direction of his motion. At that instant, his apparent weight is:

1.	25% more than his real weight.
2.	25% less than his real weight.
3.	75% more than his real weight.
4.	75% less than his real weight.

34. A block of mass 5 kg requires a minimum horizontal force of 20 N to cause it to move. Take $g = 10 \text{ m/s}^2$. Let μ_s be the coefficient of static friction, and μ_k be that of kinetic friction.



1. $\mu_s = 0.4$
2. $\mu_s = \mu_k = 0.4$
3. $\mu_k = 0.4$
4. $\mu_s \geq 0.4$

35. The upper half of an inclined plane of inclination θ is perfectly smooth while the lower half is rough. A block starting from rest at the top of the plane will again come to rest at the bottom if the coefficient of friction between the block and the lower half of the plane is given by:

1. $\mu = 2/\tan \theta$
2. $\mu = 2 \tan \theta$
3. $\mu = \tan \theta$
4. $\mu = 1/\tan \theta$

36. Consider a drop of rainwater having a mass of 1 gm falling from a height of 1 km. It hits the ground with a speed of 50 m/s. Take g as constant with a value 10 m/s^2 . The work done by the

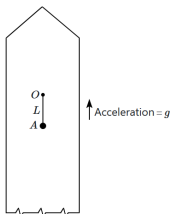
(i) gravitational force and the (ii) resistive force of air is:

1. (i) 1.25 J; (ii) - 8.25 J
2. (i) 100 J; (ii) 8.75 J
3. (i) 10 J; (ii) - 8.75 J
4. (i) - 10 J; (ii) - 8.75 J

37. A particle is displaced through $(3\hat{i} + 4\hat{j})$ m by force $2\hat{i}$ N. The work done is:

1. 14 J
2. 8 J
3. 6 J
4. 10 J

38. A simple pendulum of length L is suspended from a point O in a rocket which is ready to be launched from earth. The rocket takes off with an upward acceleration equal to g . Immediately after take off, the bob of the pendulum is given a horizontal velocity so that it just completes a vertical circle. The bob's speed (relative to the rocket) at its highest point is v_H , where:



1. $v_H = \sqrt{gL}$
2. $v_H = \sqrt{2gL}$
3. $v_H = \sqrt{3gL}$
4. $v_H = \sqrt{5gL}$

39. A block of mass 5 units is moving horizontally at speed of 1.5 ms^{-1} . A vertically upward force 5 N acts on it for 4 s. What will be the distance of the block from the point where the force starts acting after 4 s? (Ignore gravity)

1. 2 m
2. 6 m
3. 14 m
4. 10 m

40. Given below are two statements:

Assertion (A):	Work done by all forces, conservative or non-conservative, equals the change in the total energy of the system.
Reason (R):	According to Newton's laws of motion, the acceleration of any particle of the system is due to all forces acting on it; and therefore the work done on it by these is the total work done.

- Both (A) and (R) are True and (R) is the correct explanation of (A).
- Both (A) and (R) are True but (R) is not the correct explanation of (A).
- (A) is True but (R) is False.
- (A) is False but (R) is True.

41. A block of mass 1 kg moving with a velocity of 4 m/s strikes a second block of mass 2 kg at rest. The first block is brought to rest by the collision. The coefficient of restitution is:

- $\frac{1}{2}$
- $\frac{1}{3}$
- $\frac{2}{3}$
- $\frac{1}{4}$

42. A rod of mass M and length L is suspended vertically at its highest point. The rod is held so that it is horizontal and free to rotate about A and then released. There is no friction anywhere.



The force exerted by the hinge at A , when the rod is at its lowest position is:

- $2 Mg$
- $3 Mg$
- $4 Mg$
- $2.5 Mg$

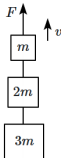
43. When an object is shot from the bottom of a long, smooth inclined plane kept at an angle of 60° with horizontal, it can travel a distance x_1 along the plane. But when the inclination is decreased to 30° and the same object is shot with the same velocity, it can travel x_2 distance. Then $x_1 : x_2$ will be:

- $1 : 2\sqrt{3}$
- $1 : \sqrt{2}$
- $\sqrt{2} : 1$
- $1 : \sqrt{3}$

44. An object is moving with uniform motion, that is, at constant speed in a straight line. There are two forces acting on the object. What can we definitely conclude from this?

- The net force is in the same direction as in which the object is moving.
- Both forces are in a direction perpendicular to the direction of the motion of the object.
- The two forces have equal magnitude but they point in opposite directions.
- None of the above may be definitely concluded.

45. Three blocks with masses m , $2m$, and $3m$ are connected by strings as shown in the figure. After an upward force F is applied on block m , the masses move upward at constant speed v . What is the net force on the block of mass $2m$? (g is the acceleration due to gravity)



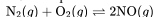
- | | | | |
|----|--------|----|--------|
| 1. | $2 mg$ | 2. | $3 mg$ |
| 3. | $6 mg$ | 4. | zero |

Chemistry

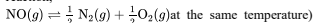
46. 50 litres of 0.1 M HCl are mixed with 50 litres of 0.2 M NaOH. The pH of the resulting solution will be:

- 12.70
- 12.34
- 8.7
- 4.2

47. The equilibrium constant for the reaction



at temperature T is 4×10^{-4} . The value of K_c for the reaction,



is :

- 2.5×10^2
- 5×10^1
- 4×10^{-4}
- 2×10^{-2}

48. The solubility product of $Cr(OH)_3$ at 298 K is 6.0×10^{-31} . The concentration of hydroxide ions in a saturated solution of $Cr(OH)_3$ will be:

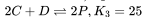
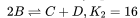
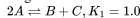
- $(2.22 \times 10^{-31})^{1/4}$
- $(18 \times 10^{-31})^{1/4}$
- $(18 \times 10^{-31})^{1/2}$
- $(4.86 \times 10^{-29})^{1/4}$

49. Consider the given two statements:

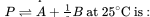
Assertion (A):	A five times increase in pressure at equilibrium doesn't change K_p for the following reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
Reason (R):	K_p depends on the degree of dissociation, not on total pressure.

1.	Both (A) and (R) are True and (R) is the correct explanation of (A).
2.	Both (A) and (R) are True but (R) is not the correct explanation of (A).
3.	(A) is True but (R) is False.
4.	(A) is False but (R) is True.

50. The equilibrium constant for the following reactions are given at $25^\circ C$:



The equilibrium constant for the reaction,



1.	$\frac{1}{20}$	2.	20
3.	$\frac{1}{42}$	4.	21

51. Given:

- K_a for acetic acid = 1.8×10^{-5}
- K_b for ammonium hydroxide = 1.8×10^{-5} (at $25^\circ C$)

Predict the nature of an aqueous solution of ammonium acetate:

- Predominantly acidic
- Predominantly basic
- Mildly acidic or mildly basic
- Perfectly neutral

52. One mole of a compound AB reacts with one mole of a compound CD according to the equation $AB(g) + CD(g) \rightleftharpoons AD(g) + CB(g)$. When equilibrium had been established, it was found that $\frac{3}{4}$ mole each of the reactants AB and CD had been converted to AD and CB. The equilibrium constant for the reaction is: (Volume remained constant during reaction)

- 6/15
- 1/9
- 19/5
- 9

53. Which of the following acids will dissociate to the greatest extent (in water)?

- HCN (hydrocyanic acid), $K_a = 6.2 \times 10^{-10}$
- HNCO (cyanic acid), $K_a = 3.3 \times 10^{-4}$
- HClO (hypochlorous acid), $K_a = 2.9 \times 10^{-8}$
- HBrO (hypobromous acid), $K_a = 2.2 \times 10^{-9}$

54. If an alcoholic beverage has a pH of 4.7, what is the concentration of OH^- ions in the solution?

(Given: $K_w = 10^{-14} \text{ mol}^2 \text{ L}^{-2}$)

- 3×10^{-10}
- 5×10^{-10}
- 1×10^{-10}
- 5×10^{-8}

55. An increase in temperature in an endothermic reversible reaction shifts the equilibrium:

- Towards the reactant's side
- Towards the product's side
- Towards neither of the sides
- Terminates the reaction

56. The concentration of Ag^+ ions in a saturated solution of $\text{Ag}_2\text{C}_2\text{O}_4$ is $2.2 \times 10^{-4} \text{ mol L}^{-1}$.

The solubility product of $\text{Ag}_2\text{C}_2\text{O}_4$ is:

1. 2.66×10^{-12}	2. 4.5×10^{-11}
3. 5.3×10^{-12}	4. 2.42×10^{-8}

57. The species, among the following, that act as both Brønsted acid and base is/are:

- H_2PO_4^-
- HPO_3^{2-}
- HPO_4^{2-}
- All of the above

58. The value of the equilibrium constant for a particular reaction is 1.6×10^{12} . When the system is in equilibrium, it will include:

- All reactants
- Mostly reactants
- Mostly products
- Similar amounts of reactants and products

59.

Assertion (A):	Salts of strong acid and strong base do not undergo hydrolysis.
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Reason (R):	These salts undergo hydration.
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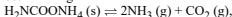
1.	Both (A) and (R) are True and (R) is the correct explanation of (A).
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2.	Both (A) and (R) are True but (R) is not the correct explanation of (A).
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3.	(A) is True but (R) is False.
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4.	Both (A) and (R) are False.
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60. For the given reaction:



Total pressure at equilibrium is found to be 18 atmospheres. The value of K_p for the above equilibrium will be:

- 72 atm³
- 144 atm³
- 432 atm³
- 864 atm³

61. The amount of water in litres that must be added to 1 litre of an aqueous solution of HCl with a pH of 1 to create an aqueous solution with pH of 2, is:

- 2.0 L
- 9.0 L
- 0.1 L
- 0.9 L

62. The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T (K).

The correct option for the pH of dimethylammonium acetate solution is:

1.	7.75	2.	6.25
3.	8.50	4.	5.50

63. If the solubility of an aqueous solution of $\text{Mg}(\text{OH})_2$ is X mol/L, what is the corresponding value of its K_{sp} ?

- $4X^3$
- $108X^5$
- $27X^4$
- $9X$

64. The decreasing order of the acidic nature of H_2SO_4 (I), H_3PO_4 (II), and HClO_4 is:

- I>II>III
- I>III>II
- III>I>II
- III>II>I

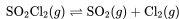
65. Pure liquids and solids are ignored while writing the expression for the equilibrium constant because:

- The size and shape of a pure substance are always fixed.
- The volume of solids and liquids is relatively fixed.
- The charges and masses of pure substances are always fixed.
- All of the above

66. The minimum volume of water required to dissolve 0.1 g lead (II) chloride to get a saturated solution (K_{sp} of $\text{PbCl}_2 = 3.2 \times 10^{-8}$; atomic mass of Pb = 207u) is:

- 1.798 L
- 0.36 L
- 17.95 L
- 0.18 L

67. At 25°C, equilibrium is established for the following reaction in a closed container:



If an inert gas, helium, is introduced into the system, which of the following statements about the behavior of the reaction is accurate?

- The concentrations of SO_2 , Cl_2 and SO_2Cl_2 do not change.
- More chlorine is formed.
- The concentration of SO_2 is reduced.
- More SO_2Cl_2 is formed.

68. Given that the pK_a values of acids A and B are 4 and 5, respectively, what can be inferred about their relative strengths?

- Acid A is 10 times stronger than acid B.
- The strength of acid A relative to acid B is in the ratio of 4:5.
- The strength of two acids cannot be compared.
- Acid B is 10 times stronger than acid A.

69. In the case of a sparingly soluble salt A_pB_q , what is the relationship between its solubility product (L_S) with its solubility (S)?

- $L_S = S^{p+q} \cdot p^p \cdot q^q$
- $L_S = S^{p+q} \cdot p^q \cdot q^p$
- $L_S = S^{pq} \cdot p^p \cdot q^q$
- $L_S = S^{pq} \cdot (pq)^{p+q}$

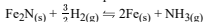
70. The K_{sp} of Ag_2CrO_4 , AgCl , AgBr , and AgI are respectively, 1.1×10^{-12} , 1.8×10^{-10} , 5.0×10^{-13} , 8.3×10^{-17} . Which one of the following salts will precipitate last if AgNO_3 solution is added to the solution containing equal moles of NaCl , NaBr , NaI , and Na_2CrO_4 ?

- AgI
- AgCl
- AgBr
- Ag_2CrO_4

71. 100 ml of HCl ($\text{pH}=1$) is mixed with 150 ml of HCl ($\text{pH}=2$). The pH of the resulting solution will be: ($\log 4.6 = 0.65$)

- 1.35
- 1.5
- 1.8
- 1.9

72. The correct expression for the following reaction is:



- $K_c = K_p(\text{RT})$
- $K_c = K_p(\text{RT})^{-3/2}$
- $K_c = K_p(\text{RT})^{-1/2}$
- $K_c = K_p(\text{RT})^{1/2}$

73. The solubility of AgCl is minimum in:

- 0.01 M NaCl
- 0.02 M CaCl_2
- 0.05 M AgNO_3
- 0.01 M NH_4OH

74. On adding inert gas at constant volume, equilibrium shifts in the forward direction will take place in -

- $\text{N}_2(g) + 3\text{H}_2(g) \rightleftharpoons 2\text{NH}_3(g)$
- $\text{PCl}_5(g) \rightleftharpoons \text{PCl}_3(g) + \text{Cl}_2(g)$
- $\text{H}_2(g) + \text{I}_2(g) \rightleftharpoons 2\text{HI}(g)$
- None of the above.

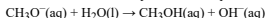
75.

Assertion (A):	A catalyst affects the final state of the equilibrium.
Reason (R):	A catalyst enables the system to attain a new equilibrium state by complexing with the reagents.

In light of the above statements choose answer from the options given below:

1.	Both (A) and (R) are True and (R) is the correct explanation of (A).
2.	Both (A) and (R) are True but (R) is not the correct explanation of (A).
3.	(A) is True but (R) is False.
4.	Both (A) and (R) are False.

76. Consider this ionic equation in an aqueous solution:



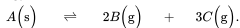
Which statement about this reaction is correct?

1.	CH_3OH is the conjugate acid of CH_3O^- .
2.	The CH_3O^- ion is a Lewis acid.
3.	The final mixture of solutions would have a pH below 7.
4.	Water is acting as a Bronsted-Lowry base.

77. For the equilibrium of the reaction, $\text{HgO}(\text{s}) \rightleftharpoons \text{Hg}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g})$, K_p for the reaction at a total pressure of "P" will be:

1. $K_p = \frac{2}{3^{3/2}} P^{3/2}$	2. $K_p = \frac{2}{3^{1/2}} P^{1/2}$
3. $K_p = \frac{1}{3^{2/3}} P^{2/3}$	4. $K_p = \frac{1}{3^{2/5}} P$

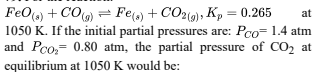
78. Given a system:



If the concentration of C at equilibrium is increased by a factor 2, it will cause the equilibrium concentration of B to change to:

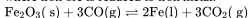
- two times of its original value
- one half of its original value
- $2\sqrt{2}$ times of its original value
- $\frac{1}{2\sqrt{2}}$ times of its original value

79. For the reaction:



1. 4.61 atm	2. 1.74 atm
3. 0.46 atm	4. 0.17 atm

80. The following reaction occurs in the Blast Furnace where iron ore is reduced to iron metal.



Using Le Chatelier's principle, predict which one of the following will not disturb the equilibrium:

- Removal of CO
- Removal of CO_2
- Addition of CO_2
- Addition of Fe_2O_3

81. The salt solution that is basic in nature is:

- Ammonium chloride.
- Ammonium sulphate.
- Ammonium nitrate.
- Sodium acetate.

82. The pH of the solution containing 50 mL each of 0.10 M sodium acetate and 0.01 M acetic acid is: [Given pK_a of $\text{CH}_3\text{COOH} = 4.57$]

- 2.57
- 5.57
- 3.57
- 4.57

83. For the reversible reaction, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ at 500°C , the value of K_p is 1.44×10^{-5} when partial pressure is measured in atmospheres. The corresponding value of K_c , with concentration in mole litre⁻¹, will be:

1. $\frac{1.44 \times 10^{-5}}{(0.082 \times 500)^{-2}}$	2. $\frac{1.44 \times 10^{-5}}{(8.314 \times 773)^{-2}}$
3. $\frac{1.44 \times 10^{-5}}{(0.082 \times 773)^2}$	4. $\frac{1.44 \times 10^{-5}}{(0.082 \times 773)^{-2}}$

84. Which one of the following is true for any diprotic acid, H_2X ?

- $K_{a1} = K_{a2}$
- $K_{a2} > K_{a1}$
- $K_{a1} > K_{a2}$
- $K_{a2} = 1/K_{a1}$

85. The ionization constant of propanoic acid is 1.32×10^{-5} . The degree of ionization of 0.05M acid solution will be:

1. $\alpha = 0.63 \times 10^{-2}$
2. $\alpha = 1.63 \times 10^{-4}$
3. $\alpha = 1.63 \times 10^{-2}$
4. $\alpha = 0.05 \times 10^{-2}$

86. Given the reaction: $A + B \rightleftharpoons C + D$. If, initially, the concentration of A and B are equal and, at equilibrium, the concentration of D is twice that of A, then what will be the equilibrium constant of a reaction?

1. 4/9
2. 9/4
3. 1/9
4. 4

87. The value of K_c of the reaction $2A \rightleftharpoons B + C$ is 2×10^{-3} . At a given time, the composition of the reaction mixture is $[A] = [B] = [C] = 3 \times 10^{-4}$ M. Select the correct option:

1. $Q_c < K_c$
2. $Q_c > K_c$
3. $Q_c = K_c$
4. $Q_c \neq K_c$

88. The correct statement(s) among the following is/are:

(I)	The equilibrium constant is independent of the initial concentration
(II)	$K > 1$ implies a spontaneous reaction
(III)	For homogeneous equilibrium, always $K_p = K_c$

1. (I) only
2. (II) and (III) only
3. (I) and (II) only
4. (I), (II) and (III)

89. The molar solubility of CaF_2 ($K_{sp} = 5.3 \times 10^{-11}$) in 0.1 M solution of NaF will be:

1. $5.3 \times 10^{-11} \text{ mol L}^{-1}$	2. $5.3 \times 10^{-8} \text{ mol L}^{-1}$
3. $5.3 \times 10^{-9} \text{ mol L}^{-1}$	4. $5.3 \times 10^{-10} \text{ mol L}^{-1}$

90. What is the solubility product of AuCl_3 , if the solubility of gold(III) chloride is $1.00 \times 10^{-4} \text{ g L}^{-1}$? [Given: molar mass = 303.3 g/mol]

1. 1.00×10^{-16}	2. 2.7×10^{-15}
3. 1.2×10^{-26}	4. 3.2×10^{-25}

Biology

91. Which of the following will not be applicable to the Gymnosperm *Cycas*?

1. Coralloid roots
2. Pinnate persistent leaves
3. Dioecy
4. Branched stems

92. Heparin:

1.	is an anti-coagulant secreted by the basophils in the blood
2.	is an anti-coagulant secreted by the eosinophils in the blood
3.	is a thrombolytic secreted by the basophils in the blood
4.	is a thrombolytic secreted by the eosinophils in the blood

93. Identify the correct statement:

1.	Monocotyledonous plants generally have taproot systems with one or more primary and secondary roots.
2.	The root hair zone develops in the region of elongation.
3.	Roots that develop from a stem, leaf, or other plant part are called adventitious roots.
4.	Apical meristem is found in the region of maturation.

94. Read the following statements and choose the set of correct statements :

A.	In cymose inflorescence, the flowers are borne in an acropetalous succession.
B.	In gulmohar, the flowers are actinomorphic.
C.	In the flowers of cucumber, the margin of the thalamus grows upward enclosing the ovary completely and getting fused with it, the other parts of flowers arise above the ovary.
D.	Bracts are reduced leaves found at the base of the pedicle.
E.	The flowers are hypogynous in mustard.

Choose the correct answer from the options given below:

1. A, B, C only
2. B, D, E only
3. A, C, D only
4. C, D, E only

95. The fungal class ascomycetes does not include:

1. *Morels and truffles*
2. *Neurospora and Claviceps*
3. *Aspergillus, Saccharomyces*
4. Puff balls and smuts

96. The primitive prokaryotes responsible for the production of biogas from the dung of ruminant animals, including the:

1. thermoacidophiles
2. methanogens
3. eubacteria
4. halophiles

97. Match each item in Column I with one in Column II and select the correct match from the codes given:

Column I		Column II	
A	Phycomycetes	P	<i>Ustilago and Puccinia</i>
B	Ascomycetes	Q	<i>Alternaria and Trichoderma</i>
C	Basidiomycetes	R	<i>Rhizopus and Albugo</i>
D	Deuteromycetes	S	<i>Aspergillus and Neurospora</i>

Codes:

	A	B	C	D
1.	R	S	P	Q
2.	R	S	Q	P
3.	S	R	P	Q
4.	S	R	Q	P

98. Match the type of nitrogenous waste with the corresponding organism group:

Column I	Column II
A. Ammonotelism	P. Birds
B. Ureotelism	Q. Mammals
C. Uricotelism	R. Aquatic animals

Codes:

	A	B	C
1.	P	Q	R
2.	Q	P	R
3.	R	P	Q
4.	R	Q	P

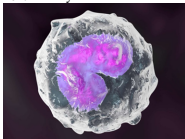
99. All the following are example of green alga except:

1. *Ectocarpus*
2. *Chara*
3. *Ulothrix*
4. *Spirogyra*

100. Potato spindle tuber disease is caused by a:

1. Viroid
2. Virion
3. Prion
4. Obelisk

101. Identify the WBC:



1. Neutrophil
2. Basophil
3. Lymphocyte
4. Monocyte

102. Which disease is caused in humans by the protozoan *Trypanosoma gambiense*?

1. Malaria
2. African sleeping sickness
3. Elephantiasis
4. Dysentery

103. *Colocasia* represents an example of

1. Underground modified root
2. Underground modified fruit
3. Underground modified stem
4. Modified adventitious root

104. Which of the following pairs correctly identifies a leaf tendril and a stem tendril?

1.	Leaf tendril: Cucumber (<i>Cucumis sativus</i>), Stem tendril: <i>Gloriosa</i> (<i>Gloriosa superba</i>).
2.	Leaf tendril: Pea (<i>Pisum sativum</i>), Stem tendril: Grapevine (<i>Vitis vinifera</i>).
3.	Leaf tendril: Citrus (<i>Citrus</i> spp.), Stem tendril: Pumpkin (<i>Cucurbita maxima</i>).
4.	Leaf tendril: Watermelon (<i>Citrullus lanatus</i>), Stem tendril: Bougainvillea (<i>Bougainvillea</i> spp.).

105. In arid regions, some plants modify their stems into flattened (*Opuntia*) or fleshy cylindrical (*Euphorbia*) structures. What are these modifications primarily for?

1. Storage of food
2. Protection from herbivores
3. Photosynthesis and water conservation
4. Climbing and support

106. Identify the correct statements regarding the inflorescence shown in the given picture:



Statement I: It is a racemose inflorescence.

Statement II: The flowers are borne in a basipetal order.

1. Only **Statement I**
2. Only **Statement II**
3. Both **Statement I** and **Statement II**
4. Neither **Statement I** nor **Statement II**

107. The maximum amount of water from the glomerular filtrate is reabsorbed in the nephron by:

1. Proximal Convoluted Tubule (PCT)
2. Distal Convoluted Tubule (DCT)
3. Loop of Henle – ascending limb
4. Loop of Henle – descending limb

108. The epithelial cells of Bowman's capsule, arranged in an intricate manner so as to leave some minute spaces called filtration slits or slit pores, are called as:

- | | |
|-----------------|--------------------------|
| 1. Macula densa | 2. Juxtaglomerular cells |
| 3. Podocytes | 4. Mesangial cells |

109. During the movement of the glomerular ultrafiltrate through the descending limb of the loop of Henle:

I: water moves into the nephron

II: solutes move out of the nephron

III: water moves out of the nephron

1. Only **I**
2. Only **II** and **III**
3. Only **III**
4. Only **I** and **II**

110. The leaves in pteridophyta are:

- | |
|---|
| 1. microphylls in <i>Selaginella</i> or macrophylls in ferns. |
| 2. macrophylls in <i>Selaginella</i> or microphylls in ferns. |
| 3. microphylls in <i>Selaginella</i> and ferns. |
| 4. macrophylls in <i>Selaginella</i> and ferns. |

111. Match each item in Column I with one in Column II and select the correct match from the codes given:

Column I	Column II
A Actinomorphic flower	P <i>Cassia</i>
B Zygomorphic flower	Q <i>Datura</i>
C Asymmetrical flower	R <i>Canna</i>

Codes:

	A	B	C
1.	P	Q	R
2.	Q	P	R
3.	Q	R	P
4.	P	R	Q

112. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A):	Juxta Glomerular Apparatus (JGA) plays an important role in regulation of glomerular filtration rate (GFR).
Reason (R):	A fall in GFR can activate the J.G. cells to release renin which can stimulate the glomerular blood flow and thereby bringing the GFR to normal level.

In the light of the above statements, choose the correct answer from the options given below:

- | | |
|----|--|
| 1. | Both (A) and (R) are True and (R) is the correct explanation of (A). |
| 2. | Both (A) and (R) are True but (R) is not the correct explanation of (A). |
| 3. | (A) is True but (R) is False. |
| 4. | (A) is False but (R) is True. |

113. Regarding the heart sounds heard during a cardiac cycle:

1.	The first heart sound is produced due to simultaneous closure of the tricuspid and the bicuspid valves.
2.	The second heart sound is produced by the simultaneous closure of the aortic and the pulmonic semilunar valves.
3.	Both 1 and 2 are correct.
4.	Both 1 and 2 are incorrect

114. Which evolutionary advancement in angiosperms allows for the efficient transport of water, nutrients, and food, contributing to their dominance among terrestrial plants?

1. Presence of spores
2. Development of seeds
3. Complex vascular system
4. Formation of cones

115. Which metabolic disorder is indicated by the presence of glucose in urine (glycosuria) and presence of ketone bodies in urine [ketonuria]?

1. Uremia	2. Diabetes mellitus
3. Glomerulonephritis	4. Renal calculi

116. What makes Red Algae different from Green Algae and Brown Algae?

1.	Unlike Green Algae and Brown Algae, Red Algae do not have chlorophyll a.
2.	Unlike Green Algae and Brown Algae, Red Algae have no differentiated cells.
3.	Unlike Green Algae and Brown Algae, Red Algae are not photosynthetic.
4.	Unlike Green Algae and Brown Algae, Red Algae do not have any flagellated stages in their life cycles.

117. An example of colonial algae is:

1. <i>Volvox</i>	2. <i>Ulothrix</i>
3. <i>Spirogyra</i>	4. <i>Chlorella</i>

118. During the ventricular systole:

I:	The atria also undergo systole
II:	The atrioventricular valves get closed
III:	The semilunar valves are forced open
1.	Only I and II are correct
2.	Only I and III are correct
3.	Only II and III are correct
4.	I, II and III are correct

119. All the following will be true for mycoplasma except:

1.	They completely lack a cell wall.
2.	They are the smallest living cells known.
3.	They can survive without oxygen.
4.	They are unable to cause disease in animals and plants.

120. In China Rose:

A: Aestivation is called as twisted

B: Stamens are monadelphous

C: Placentation is axile.

1. Only A and B are correct.
2. Only A and C are correct.
3. Only B and C are correct.
4. A, B, and C are correct.

121. What is the main function of the sebaceous glands in the excretory system?

1.	To excrete nitrogenous wastes through sweat
2.	To release excess salts and urea into the blood
3.	To secrete oily substances like sterols and waxes to protect the skin
4.	To release glucose and lactic acid into the bloodstream

122. During the blood clotting process, what is the correct sequence of events involving thrombin and fibrin?

1.	Thrombin activates fibrinogen to form fibrin, which traps blood cells.
2.	Thrombin forms from prothrombin and breaks down fibrin into fibrinogen.
3.	Fibrinogen activates thrombin, which stabilizes the clot.
4.	Thrombin combines with fibrinogen to form inactive fibrin, preventing further clotting.

123. Cyanobacteria:**I:** belong to Kingdom Monera.**II:** are photosynthetic autotrophs.

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

124. Consider the given statements:**I:** Prop roots are seen in Banyan tree**II:** Stilt roots are seen in maize and sugarcane**III:** Pneumatophores are seen in *Rhizophora*

1. Only **I** and **II** are correct
2. Only **I** and **III** are correct
3. Only **II** and **III** are correct
4. **I, II** and **III** are correct

125. Match List-I with List-II:

List-I		List-II
A. Tricuspid valves	I.	Guards the opening between right atrium and right ventricle
B. Sino-atrial node	II.	Guards the opening between the left atrium and the left ventricle
C. Mitral valves	III.	Guards the opening of right and left ventricles into pulmonary artery and aorta, respectively
D. Semilunar valves	IV.	Cardiac musculature in the right upper corner of right atrium

Choose the correct answer from the options given below:

1. A-III, B-IV, C-II, D-I
2. A-II, B-I, C-IV, D-III
3. A-III, B-IV, C-I, D-II
4. A-I, B-IV, C-II, D-III

126. Mosses reproduce by vegetative means through:

1. Fragmentation and budding in secondary protonema.
2. Only budding in secondary protonema but not by fragmentation in secondary protonema.
3. By adventitious root formation and budding in secondary protonema.
4. By fragmentation and budding in the sporophyte.

127. The most abundant prokaryotes helpful to human in making curd from milk and in production of antibiotics are the ones categorized as:

1. Cyanobacteria
2. Archaeobacteria
3. Chemosynthetic autotrophs
4. Heterotrophic bacteria

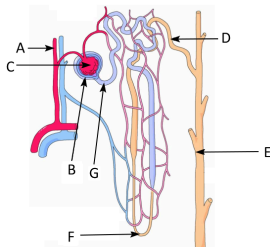
128. Phylogenetic classification systems are based on:

1. all observable characteristics.
2. cytological information like chromosome number, structure, behaviour.
3. evolutionary relationships between the various organisms.
4. natural affinities among the organisms and consider, not only the external features, but also internal features, like ultra-structure, anatomy, embryology and phytochemistry.

129. The fruit in mango and coconut is called as:

1. Aril
2. Drupe
3. Berry
4. Samara

130. How many of the given statements regarding the given figure are correct?



I:	A is the afferent arteriole, B is Bowman's Capsule and C is efferent arteriole.
II:	D is lined with cuboidal brush bordered epithelium and G is lined with simple cuboidal epithelium
III:	At F, the filtrate is most dilute and at G the filtrate is most concentrated

- 0
- 1
- 2
- 3

131. How many of the given statements regarding leaves of angiosperms are true?

I:	Leaves originate from shoot apical meristems and are arranged in a basipetal order
II:	Pulvinus is seen in some leguminous plants
III:	Parallel venation is characteristic of most dicots
IV:	A bud is present in the axil of petiole of both simple and compound leaves
V:	Palmately compound leaves are seen in neem

- 5
- 4
- 3
- 2

132. Roots in gymnosperms exhibit special associations. Pinus forms a symbiotic relationship with fungi called mycorrhiza, whereas in Cycas, special coralloid roots associate with:

1.	Rhizobium bacteria
2.	Fungi for nitrogen fixation
3.	Cyanobacteria for nitrogen fixation
4.	Archaea for nitrogen fixation

133. Viroid is an infectious agent that:

- is larger than viruses.
- was found to be a free DNA.
- lacks the protein coat that is found in viruses.
- has RNA of a very high molecular weight.

134. Consider the given two statements:

Statement I:	An increase in blood flow to the atria of the heart can cause the release of Atrial Natriuretic Factor (ANF).
Statement II:	A fall in glomerular blood flow/glomerular blood pressure/GFR can activate the JG cells to release renin.

- Statement I** is correct; **Statement II** is incorrect
- Statement I** is correct; **Statement II** is correct
- Statement I** is incorrect; **Statement II** is incorrect
- Statement I** is incorrect; **Statement II** is correct

135. What is the function of the coronary arteries?

- Supply oxygenated blood to the lungs
- Deliver deoxygenated blood to the liver
- Transport hormones to the heart
- Supply oxygenated blood to the heart muscle

136. Regarding erythroblastosis fetalis:

I:	The baby is Rh positive and the mother is Rh negative.
II:	The mother's immune system has made antibodies against the Rh factor present on the red blood cells of the previous baby.
III:	Anaemia and jaundice in a baby are rare.
IV:	There is no mechanism to prevent its occurrence.

- I, II, III** and **IV** are correct
- Only **I** and **II** are correct
- Only **III** and **IV** are correct
- Only **I** is correct

137. Identify the incorrect statements regarding phycomycetes:

I.	Found in aquatic habitats and on decaying wood in moist and damp places or as obligate parasites on plants.
II.	Mycelium is aseptate and coenocytic.
III.	Asexual reproduction takes place by zoospores (motile) or by aplanospores (non-motile).
IV.	These spores are endogenously produced in sporangium.
V.	A zygospore is formed by the fusion of two gametes.

1. None, all are correct
2. Only III
3. Only II, III, IV and V
4. Only I

138. Read the following statements and choose the set of correct statements:

In the members of Phaeophyceae,

A:	Asexual reproduction occurs usually by biflagellate zoospores.
B:	Sexual reproduction is by oogamous method only.
C:	Stored food is in the form of carbohydrates which is either mannitol or laminarin.
D:	The major pigments found are chlorophyll a, c and carotenoids and xanthophyll.
E:	Vegetative cells have a cellulosic wall, usually covered on the outside by gelatinous coating of algin.

Choose the correct answer from the options given below:

1. B, C, D and E only
2. A, C, D and E only
3. A, B, C and E only
4. A, B, C and D only

139. What type of nutrition do the majority of bacteria exhibit?

1. Photosynthetic autotrophic
2. Chemosynthetic autotrophic
3. Saprophytic
4. Heterotrophic

140. The flower shown in the given figure most likely is of a plant that belongs to the family:



1. Fabaceae	2. Liliaceae
3. Solanaceae	4. Cruciferae

141. Which of the following is not a feature of mosses?

1.	Gametophyte has two stages: protonema and leafy stage.
2.	Gametophyte is dominant and bears sex organs.
3.	Sporophyte is free-living and independent of the gametophyte.
4.	Rhizoids are multicellular and branched.

142. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A):	Rh incompatibility related complication does not arise in first pregnancy even if an Rh negative mother is carrying an Rh positive foetus.
Reason (R):	Rh antigens are proteins.

In the light of the above statements choose the correct answer from the options given below:

1.	Both (A) and (R) are True and (R) is the correct explanation of (A).
2.	Both (A) and (R) are True but (R) is not the correct explanation of (A).
3.	(A) is True but (R) is False.
4.	Both (A) and (R) are False.

143. In a cardiac cycle, joint diastole state correctly means:

1. only the two atria are in a relaxed state
2. only the two ventricles are in a relaxed state
3. ventricles and atria are in a relaxed state
4. only the left atrium and the left ventricle are in a relaxed state

144. JGA is a special sensitive region formed by cellular modifications in the:

1.	DCT and the afferent arteriole at the location of their contact
2.	DCT and the efferent arteriole at the location of their contact
3.	PCT and the afferent arteriole at the location of their contact
4.	PCT and the efferent arteriole at the location of their contact

145. Biflagellate zoospores that are pear-shaped and have two unequal laterally attached flagella are involved in:

1.	Asexual reproduction in brown algae
2.	Asexual reproduction in red algae
3.	Sexual reproduction in brown algae
4.	Sexual reproduction in red algae

146. Which structure is involved in the reproduction of gymnosperms?

1. Flowers
2. Cones
3. Sporangia
4. Sori

147. A dominant gametophyte will be seen in the life cycle of:

1. *Marchantia*
2. *Sequoia*
3. *Adiantum*
4. *Eucalyptus*

148. ADH:

I:	makes the collecting duct completely impermeable to water.
II:	causes vasoconstriction.

1. Only **I** is correct
2. Only **II** is correct
3. Neither **I** nor **II** are correct
4. Both **I** and **II** are correct

149. The sinoatrial node (SAN) in the human heart is present in:

1. the right upper corner of the right atrium.
2. the right atrium near the base of inter-atrial septum.
3. the left lower corner of the left atrium.
4. the left atrium near the base of inter-atrial septum.

150. In the kingdom Monera, bacteria are classified under which major group based on their shape?

1.	Archaeobacteria, Eubacteria, Cyanobacteria, and Mycoplasma
2.	Fungi, Algae, Protozoa, and Bacteria
3.	Eukaryotes, Prokaryotes, Autotrophs, and Heterotrophs
4.	Cocci, Bacilli, Vibrio, and Spirilla

151. How many of the given statements are correct?

I:	Mosses and liverworts are examples of Bryophytes.
II:	Pteridophytes include horsetails and ferns.
III:	The plant body of angiosperms is differentiated into roots, stems, and leaves.
IV:	Algae can be found in both freshwater and marine habitats.

1. 1
2. 2
3. 3
4. 4

152. On an average, what amount of urea is excreted out per day in humans?

1. 15 – 40 mg
2. 70 – 100 mg
3. 2 – 3 gm
4. 25 – 30 gm

153. An example of a plant exhibiting whorled phyllotaxy will be:

1.	China rose	2.	Sunflower
3.	<i>Alstonia</i>	4.	Mustard

154. Which pigment is predominantly found in red algae and contributes to their distinctive coloration?

1. Chlorophyll a
2. Phycoerythrin
3. Carotenoids
4. Xanthophylls

155. Who proposed the five kingdom classification?

1. Carolus Linnaeus
2. Aristotle
3. Bentham and Hooker
4. R. H. Whittaker

156. Match the items given in **Column-I** with those in **Column-II** and select the correct option given below:

Column-I	Column-II
a. Fibrinogen	i. Osmotic balance
b. Globulin	ii. Blood clotting
c. Albumin	iii. Defense mechanism

	a	b	c
1.	iii	ii	i
2.	i	ii	iii
3.	i	iii	ii
4.	ii	iii	i

157. Which part of the nephron is responsible for reabsorption of the maximum amount of water from the filtrate?

1. Proximal convoluted tubule (PCT)
2. Distal convoluted tubule (DCT)
3. Loop of Henle
4. Collecting duct

158. A certain road accident patient with an unknown blood group needs an immediate blood transfusion. His one doctor friend at once offers his blood. What was the blood group of the donor?

1. Blood group B
2. Blood group AB
3. Blood group O
4. Blood group A

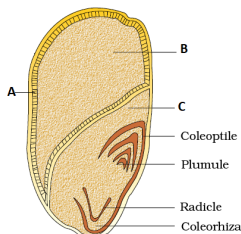
159. Identify the incorrect statement regarding viroids:

1. Viroids are the smallest infectious pathogens known.
2. They are composed solely of a short strand of circular, single-stranded RNA that has no protein coating.
3. The first recognized viroid, the pathogenic agent of the potato spindle tuber disease, was discovered by Theodor Otto Diener.
4. The nucleic acid of viroids codes for only a few proteins.

160. Identify the incorrectly matched pair:

1. Pneumatophores	<i>Rhizophora</i>
2. Leaf tendrils	Gourds
3. Stilt roots	Maize
4. Palmately compound leaves	Silk cotton

161. Identify A, B, and C in the given figure:



A	B	C
1. Aleurone layer	Scutellum	Endosperm
2. Scutellum	Endosperm	Aleurone layer
3. Aleurone layer	Endosperm	Scutellum
4. Endosperm	Scutellum	Aleurone layer

162. In which of the following groups of organisms do you expect to find the greatest metabolic diversity?

1. Fungi
2. Plants
3. Bacteria
4. Animals

163. Consider the given two statements.

Statement I:	Basophils are the cells involved in inflammatory reactions.
Statement II:	Neutrophils and monocytes are main phagocytic leukocytes.

1. Only **Statement I** is correct.
2. Only **Statement II** is correct.
3. Both **Statement I** and **Statement II** are correct.
4. Both **Statement I** and **Statement II** are incorrect.

164. Blood pressure in the mammalian aorta is maximum during:

1. systole of the left atrium
2. diastole of the right ventricle
3. systole of the left ventricle
4. diastole of the right atrium

165. Gemmae are:

1. Asexual buds found in liverworts
2. Sex organs in mosses
3. Resting zygote in many fungi
4. Endospores of certain bacteria

166. Classification system based on information like chromosome number, structure, behaviour is called as:

1. Numerical taxonomy
2. Cytotaxonomy
3. Chemotaxonomy
4. Cladistics

167. How many of the given statements are true?

A:	When stamens are attached to the petals, they are epipetalous as in brinjal
B:	When stamens are attached to the perianth, they are epiphyllous as in lily
C:	The stamens may be united into one bunch or one bundle (monadelphous) as in China rose
D:	There may be a variation in the length of filaments within a flower, as in <i>Salvia</i> and mustard

1. 1
2. 2
3. 3
4. 4

168. Match each item in Column I with one in Column II and select the correct match from the codes given below:

Column I	Column II
A <i>Ashwagandha</i>	P Ornamental
B <i>Petunia</i>	Q Dilation of pupil
C Fumigatory	R Medicine
D <i>Atropa</i>	S Tobacco

Codes:

	A	B	C	D
1.	R	P	S	Q
2.	R	S	P	Q
3.	P	R	S	Q
4.	Q	P	R	S

169. What would happen to urine output if the renal tubules lost the ability to actively transport sodium?

1.	Urine output would increase due to reduced water reabsorption.
2.	Urine output would decrease as water follows passive sodium reabsorption.
3.	The osmolarity of urine would increase drastically.
4.	Urine production would cease completely.

170. The osmolarity of interstitial fluid in renal cortex of vertebrates is about _____ mOsm per litre, which can be increased to about _____ mOsm per litre in the inner medulla during concentration of urine by the kidneys.

1. 1200, 300	2. 300, 1200
3. 300, 600	4. 600, 300

171. The red algae reproduce:

1.	asexually by non motile spores and sexually by non motile gametes
2.	asexually by non motile spores and sexually by motile gametes
3.	asexually by motile spores and sexually by non motile gametes
4.	asexually by motile spores and sexually by motile gametes

172. The clotting of blood is initiated by:

1.	The exposure of thrombokinase leading to formation of the prothrombin activator complex.
2.	The release of heparin.
3.	The destruction of red blood cells.
4.	The activation of the complement system.

173. Which of the following mechanisms is primarily responsible for the concentration of urine in the kidney?

1.	The active transport of glucose and amino acids out of the filtrate in the proximal convoluted tubule increases the filtrate's concentration.
2.	The passive reabsorption of sodium ions in the descending limb of the loop of Henle concentrates the urine.
3.	The active reabsorption of ions in the thick ascending limb of the loop of Henle, followed by the reabsorption of water in the collecting ducts in response to antidiuretic hormone (ADH), concentrates the urine.
4.	The filtration of blood under high pressure in the glomerulus results in a concentrated urine filtrate from the beginning.

174. What is the name of the type of seed where the endosperm is not completely used up during seed development, and remains in the mature seed, as seen in Castor?

1.	Exalbuminous seed	2.	Non-endospermic seed
3.	Albuminous seed	4.	Perispermic seed

175. How many of the following statements about the human heart are correct?

I:	The tricuspid valve is present between the left atrium and left ventricle.
II:	Purkinje fibers help in conducting the cardiac impulse to the myocardium of ventricles.
III:	The sinoatrial node acts as the pacemaker of the heart.
IV:	Atrial contraction increases the ventricular filling.

- One
- Two
- Three
- Four

176. Consider the given two statements:

Assertion (A):	Terrestrial adaptation necessitated the production of lesser toxic nitrogenous wastes like urea and uric acid.
Reason (R):	Terrestrial organisms must conserve water.

- (A) is True but (R) is False.
- Both (A) and (R) are True but (R) does not correctly explain (A).
- Both (A) and (R) are True and (R) correctly explains (A).
- (A) is False but (R) is True.

177. Which of the following fungi is used extensively in biochemical and genetic work?

1. Penicillium	2. Aspergillus
3. Claviceps	4. Neurospora

178. Select the wrong statement:

1.	The walls of diatoms are easily destructible
2.	Diatomaceous earth is formed by the cell walls of diatoms
3.	Diatoms are chief producers in the oceans
4.	Diatoms are microscopic and float passively in water

179. Which of the following correctly describes the cardiac output?

1.	The amount of blood pumped by both ventricles combined in one heartbeat.
2.	The volume of blood pumped by each ventricle per minute.
3.	The amount of blood ejected by each atrium per minute.
4.	The sum of the volumes of oxygenated and deoxygenated blood in the body.

180. Consider the given two statements:>

Statement I:	The filtrate becomes progressively concentrated as it moves up the ascending limb of Henle's loop and diluted as it moves down the ascending limb.
Statement II:	The descending limb is impermeable to water but permeable to electrolytes, while the ascending limb is permeable to water but impermeable to electrolytes.

Choose the correct option:

- | |
|---|
| 1. Both Statement I and Statement II are correct. |
| 2. Both Statement I and Statement II are incorrect. |
| 3. Statement I is correct, but Statement II is incorrect. |
| 4. Statement I is incorrect, but Statement II is correct. |



Mitoprep

Sapno Ki Udaan